[Edit: 1/9 - Course will be online until the end of January. Office hours will be on Zoom at https://washington.zoom.us/my/pisan]

Instructor: Dr. Yusuf Pisan, pisan@uw.edu UW1-260Q (425) 352-3741

Class: M/W 11:00-1:00pm. First week on Zoom, rest in UW1-060

Office Hours: Monday 1-2pm and Thursday 1-2pm. Signup via Canvas Calendar

Class Discord: invite link

Course Description

Transition from basic programming skills to a rigorous process of software development. Familiarization with higher level programming techniques (recursion, generic programming, stacks, queues, trees, searching, and sorting). Emphasizes connection between algorithmic thought and implementation. Engineering applications are emphasized. Prerequisite: CSS 132; CSSSKL 133 is a co-requisite.

Learning Objectives

- Use higher level programming techniques (object-orientation, recursion, inheritance, generic programming), constructs (lists, stacks, queues, trees) and algorithms (searching, sorting).
- Analyze the running time of algorithms.
- Solve programming problems that require multi-step approaches or must be decomposed into components.

Textbooks

zyBooks: Programming in C++, F Vahid and R Lysecky, https://www.zybooks.com/catalog/programming-c-plus-plus/

This is an e-book with integrated exercises. To get the e-book:

1. Sign in or create an account at https://learn.zybooks.com/ with your @uw.edu email

2. Enter zyBook code: UWBCSS133PisanWinter2022

3. Subscribe

Bradley, Aaron R., Programming for Engineers: A Foundational Approach to Learning C and Matlab, 2011, ISBN 978-3642233029. Available as a PDF through the University of Washington library
http://alliance-primo.hosted.exlibrisgroup.com/UW:all:CP71129050430001451

Grading

- Exercises: 15%

- Projects: 30%

- Exams: 35%

- Final: 20%

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. A student who achieves 75% of the possible points will receive a 2.0 grade in this course. Students with 95% and above will receive a 4.0 grade. Scores in between will be interpolated.

Assignment Submission: Exercises will be graded as Complete/Incomplete. You can miss up to 3 exercises without penalty.

For projects, you can submit one and only one project 24 hours late without any penalty. If you are using your 24 hour extension, post a brief explanation to "Using Extension" assignment.

Other than when working in pairs, talking about code is OK, looking at each others code is not OK. Looking at references to understand how a functions gets used is OK; looking up assignment solutions is not OK. It is also not acceptable for your code or solutions to be publicly accessible on the web (for example, a public GitHub repository). Plagiarism will result in an assignment score of zero and a misconduct letter in your student record. Please be very careful to adhere to the student code of conduct: http://www.washington.edu/cssc/for-students/student-code-of-conduct/ I will make allowances for exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

Attendance: Attend all classes. You are responsible for all the material covered in class, as well as any announcements including change of due dates or assignment specifications. There will also be graded in-class group exercises to practice problem solving. If you miss a class, I expect you to make-up for it on your own by asking your friends, reviewing the textbook, lecture materials, etc.

Communication: We will use discord as an extension of the classroom. Use a meaningful nickname and act professionally. If your question can be answered publicly by me or by a classmate, post it to discord. Use the office hours for complex issues or topics you are struggling with.

Use your UW email rather than "Canvas Messaging" to communicate directly with me. "Canvas Submission Comments" should only be used to draw the grader's attention to a specific part of your submission.

Problems: If you are having difficulties, come and talk to me. If I don’t know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Course Material: Lecture notes and other material will be posted to Canvas under "Files".

See the School of STEM Course Policies (Links to an external site.), which covers:

- Academic Integrity
- Access and Accommodations
- Classroom Emergency Preparedness
- For Our Veterans
- Grade of Incomplete
- Inclement Weather
- Parenting Resources
- Religious Accommodations
- Respect for Diversity

- Student Support Services

- Surviving Sexual and Relationship Violence

- Wonder How to Address Faculty?

Course Calendar

Week	Tuesday/Thursday	Notes
3 Jan 5 Jan	Introduction - C and Unix	H01
10 Jan 12 Jan	Arrays, Strings, structs, define	H02
17 Jan 19 Jan	No-Class **Exam-1**	18 Jan, Martin Luther King Day H03
24 Jan 26 Jan	Objects and Classes	H04
31 Jan 2 Feb	Streams - Queue - Stack	H05
7 Feb 9 Feb	Matlab **Exam-2**	H06

14 Feb 16 Feb	Inheritance	15 Feb, Presidents' Day H07
21 Feb 23 Feb	Recursion	H08
28 Mar 2 Mar	STL Containers **Exam-3**	H09
7 Mar 9 Mar	Search and Sort - Trees	H10
14 Mar 16 Mar	**Final** - 16 Mar	No classes during finals week

Course Summary: